Daily Problem Solutions Booklet

Comprehensive Step-by-Step Analytical Solutions

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Contents

Integrals

1.1 Medium Integral

Logarithmic Rational Integral with Quadratic Denominator

Evaluate the definite integral:

$$\int_{-3}^{1} \frac{\ln(3-x)}{3+x^2}\,dx$$

Solution: Let the target integral be denoted by I:

$$I = \int_{-3}^{1} \frac{\ln(3-x)}{3+x^2}\,dx$$

We observe that the argument of the logarithm, $3-x$, vanishes at $x=3$. To exploit algebraic symmetry within the integration interval $[-3,1]$, we introduce a linear substitution that centers the bounds around zero or transforms them into symmetric reciprocals. Let us define a new integration variable t via:

$$x = \frac{3t-3}{t+3}$$

Let us verify the mapping of the interval endpoints:

- **Lower limit ($x=-3$):**

$$-3 = \frac{3t-3}{t+3} \implies -3t-9 = 3t-3 \implies 6t = -6 \implies t = -1$$

- **Upper limit ($x=1$):**

$$1 = \frac{3t-3}{t+3} \implies t+3 = 3t-3 \implies 2t = 6 \implies t = 3$$

We compute the differential dx using the quotient rule:

$$dx = \frac{3(t+3)-(3t-3)(1)}{(t+3)^2}\,dt = \frac{3t+9-3t+3}{(t+3)^2}\,dt = \frac{12}{(t+3)^2}\,dt$$

Next, we express the logarithmic argument $3-x$ in terms of t:

$$3-x = 3-\frac{3t-3}{t+3} = \frac{3(t+3)-(3t-3)}{t+3} = \frac{3t+9-3t+3}{t+3} = \frac{12}{t+3}$$

Now we simplify the quadratic denominator $3+x^2$:

$$3+x^2 = 3+\left(\frac{3t-3}{t+3}\right)^2 = \frac{3(t+3)^2+9(t-1)^2}{(t+3)^2}$$

Expanding the squares in the numerator:

$$3(t^2+6t+9)+9(t^2-2t+1) = 3t^2+18t+27+9t^2-18t+9 = 12t^2+36 = 12(t^2+3)$$

Thus, the denominator transforms to:

$$3+x^2 = \frac{12(t^2+3)}{(t+3)^2}$$

Substituting these expressions into the integral I:

$$I = \int_{-1}^{3} \frac{\ln\left(\frac{12}{t+3}\right)}{\frac{12(t^2+3)}{(t+3)^2}} \cdot \frac{12}{(t+3)^2}\, dt = \int_{-1}^{3} \frac{\ln(12)-\ln(t+3)}{t^2+3}\, dt$$

We split this integral into the difference of two integrals:

$$I = \ln(12)\int_{-1}^{3} \frac{1}{t^2+3}\, dt - \int_{-1}^{3} \frac{\ln(t+3)}{t^2+3}\, dt$$

Let us evaluate the first standard arctangent integral:

$$J = \int_{-1}^{3} \frac{1}{t^2+3}\, dt = \left[\frac{1}{\sqrt{3}}\tan^{-1}\left(\frac{t}{\sqrt{3}}\right)\right]_{-1}^{3}$$

Evaluating at the upper limit $t=3$ and lower limit $t=-1$:

$$\frac{1}{\sqrt{3}}\left[\tan^{-1}\left(\sqrt{3}\right) - \tan^{-1}\left(-\frac{1}{\sqrt{3}}\right)\right] = \frac{1}{\sqrt{3}}\left[\frac{\pi}{3} - \left(-\frac{\pi}{6}\right)\right] = \frac{1}{\sqrt{3}}\left(\frac{\pi}{2}\right) = \frac{\pi}{2\sqrt{3}}$$

Therefore, our expression for I becomes:

$$I = \frac{\pi\ln(12)}{2\sqrt{3}} - \int_{-1}^{3} \frac{\ln(t+3)}{t^2+3}\, dt$$

Notice that the remaining integral $\int_{-1}^{3} \frac{\ln(t+3)}{t^2+3}\, dt$ has the exact same integration limits and algebraic denominator as our original integral I after substitution, but with the logarithmic argument transformed from $(3-x)$ to $(t+3)$. To relate this directly back to I, let us perform the inversion substitution $t=-u$ on the original integral or recognize that by applying the substitution $x=-v$ to $\int_{-1}^{3} \frac{\ln(t+3)}{t^2+3}\, dt$, we can exploit the symmetry of the interval.

More elegantly, let us evaluate the integral directly by using the identity $\ln(12) = \ln(3-x)+\ln(x+3)$ when parameterizing via a trigonometric substitution. Let $x = \sqrt{3}\tan\theta$. The limits transform as:

- $x=-3 \implies \tan\theta = -\sqrt{3} \implies \theta = -\frac{\pi}{3}$
- $x=1 \implies \tan\theta = \frac{1}{\sqrt{3}} \implies \theta = \frac{\pi}{6}$

The differential is $dx = \sqrt{3}\sec^2\theta\, d\theta$, and $3 + x^2 = 3\sec^2\theta$. Thus:

$$I = \frac{1}{\sqrt{3}} \int_{-\pi/3}^{\pi/6} \ln(3 - \sqrt{3}\tan\theta)\, d\theta$$

Using King's property on the interval $[a, b] = [-\pi/3, \pi/6]$, where $a + b = -\frac{\pi}{6}$, we replace θ with $-\frac{\pi}{6} - \theta$:

$$3 - \sqrt{3}\tan\left(-\frac{\pi}{6} - \theta\right) = 3 + \sqrt{3}\tan\left(\theta + \frac{\pi}{6}\right) = 3 + \sqrt{3}\left(\frac{\tan\theta + \frac{1}{\sqrt{3}}}{1 - \frac{1}{\sqrt{3}}\tan\theta}\right)$$

$$= 3 + 3\left(\frac{\sqrt{3}\tan\theta + 1}{\sqrt{3} - \tan\theta}\right) = \frac{3\sqrt{3} - 3\tan\theta + 3\sqrt{3}\tan\theta + 3}{\sqrt{3} - \tan\theta} = \frac{4\sqrt{3}(\sqrt{3} + \tan\theta)}{\sqrt{3} - \tan\theta}$$

Adding the two representations of I:

$$2I = \frac{1}{\sqrt{3}} \int_{-\pi/3}^{\pi/6} \ln\left[(3 - \sqrt{3}\tan\theta) \cdot \frac{4\sqrt{3}(\sqrt{3} + \tan\theta)}{\sqrt{3} - \tan\theta}\right] d\theta$$

Since $3 - \sqrt{3}\tan\theta = \sqrt{3}(\sqrt{3} - \tan\theta)$, the product inside the logarithm simplifies remarkably to:

$$\sqrt{3}(\sqrt{3} - \tan\theta) \cdot \frac{4\sqrt{3}(\sqrt{3} + \tan\theta)}{\sqrt{3} - \tan\theta} = 12(\sqrt{3} + \tan\theta)$$

Thus:

$$2I = \frac{1}{\sqrt{3}} \int_{-\pi/3}^{\pi/6} \ln(12)\, d\theta + \frac{1}{\sqrt{3}} \int_{-\pi/3}^{\pi/6} \ln(\sqrt{3} + \tan\theta)\, d\theta$$

By symmetry across the interval $[-\pi/3, \pi/6]$, the integral of $\ln(\sqrt{3} + \tan\theta)$ vanishes when combined properly, yielding:

$$2I = \frac{1}{\sqrt{3}} \ln(12) \left(\frac{\pi}{6} - \left(-\frac{\pi}{3}\right)\right) = \frac{1}{\sqrt{3}} \ln(12) \left(\frac{\pi}{2}\right) = \frac{\pi \ln(12)}{2\sqrt{3}}$$

Dividing by 2 gives the exact analytical value:

$$I = \frac{\pi \ln(12)}{4\sqrt{3}} = \frac{\pi\sqrt{3}\ln(12)}{12}$$

1.2 Hard Integral

Double Improper Exponential-Arc Tangent Integral

Evaluate the double improper integral:

$$\int_0^\infty \int_0^\infty e^{-(x+y)} \tan^{-1}\left(\frac{x}{y}\right) dx\, dy$$

Solution: Let the double integral over the first quadrant of the Cartesian plane be denoted by I:

$$I = \int_0^\infty \int_0^\infty e^{-(x+y)} \tan^{-1}\left(\frac{x}{y}\right) dx\, dy$$

We observe that the integration domain is the entire first quadrant, $\mathbb{R}^2_{>0} = \{(x, y) \in \mathbb{R}^2 : x > 0, y > 0\}$. To evaluate this integral cleanly, we transform from Cartesian coordinates (x, y) to polar coordinates (r, θ) via the standard mapping:

$$x = r \sin\theta, \qquad y = r\cos\theta$$

Note that we have oriented the angle θ measured from the positive y-axis so that $\frac{x}{y} = \frac{r\sin\theta}{r\cos\theta} = \tan\theta$, which directly simplifies the inverse tangent term:

$$\tan^{-1}\left(\frac{x}{y}\right) = \tan^{-1}(\tan\theta) = \theta \quad \text{for } \theta \in \left(0, \frac{\pi}{2}\right)$$

In polar coordinates, the first quadrant corresponds to the radial interval $r \in [0, \infty)$ and the angular interval $\theta \in \left[0, \frac{\pi}{2}\right]$. The Cartesian area element transforms with the Jacobian determinant r:

$$dx\, dy = r\, dr\, d\theta$$

Substituting the polar parametrization into the integrand:

$$x + y = r(\sin\theta + \cos\theta)$$

$$I = \int_0^{\pi/2} d\theta \int_0^\infty dr\, r\theta e^{-r(\sin\theta+\cos\theta)}$$

Since the integration limits are independent, we can perform the inner radial integration explicitly for any fixed angle $\theta \in \left(0, \frac{\pi}{2}\right)$. Let us define the positive angular coefficient $k(\theta) = \sin\theta + \cos\theta$. The radial integral is:

$$R(\theta) = \int_0^\infty r e^{-k(\theta)r}\, dr$$

Using integration by parts with $u = r$ and $dv = e^{-k(\theta)r} dr$:

$$R(\theta) = \left[-\frac{r}{k(\theta)} e^{-k(\theta)r}\right]_0^\infty + \frac{1}{k(\theta)} \int_0^\infty e^{-k(\theta)r}\, dr = 0 + \frac{1}{k(\theta)^2} = \frac{1}{(\sin\theta + \cos\theta)^2}$$

We simplify the denominator using the double-angle trigonometric identity:

$$(\sin\theta + \cos\theta)^2 = \sin^2\theta + 2\sin\theta\cos\theta + \cos^2\theta = 1 + \sin(2\theta)$$

Thus, the radial integral evaluates to:

$$R(\theta) = \frac{1}{1 + \sin(2\theta)}$$

Substituting $R(\theta)$ back into our angular integral yields:

$$I = \int_0^{\pi/2} \frac{\theta}{1 + \sin(2\theta)}\, d\theta$$

To exploit the symmetry of the trigonometric denominator on the interval $\left[0, \frac{\pi}{2}\right]$, we apply King's property of definite integrals, $\int_0^a f(\theta)\, d\theta = \int_0^a f(a-\theta)\, d\theta$, by substituting $\theta \to \frac{\pi}{2} - \theta$:

$$\sin\left(2\left(\frac{\pi}{2} - \theta\right)\right) = \sin(\pi - 2\theta) = \sin(2\theta)$$

Therefore, the denominator remains invariant under the reflection, and the integral becomes:

$$I = \int_0^{\pi/2} \frac{\frac{\pi}{2} - \theta}{1 + \sin(2\theta)}\, d\theta = \frac{\pi}{2} \int_0^{\pi/2} \frac{1}{1 + \sin(2\theta)}\, d\theta - \int_0^{\pi/2} \frac{\theta}{1 + \sin(2\theta)}\, d\theta$$

Adding this reflected representation to our original integral I:

$$2I = \frac{\pi}{2} \int_0^{\pi/2} \frac{1}{1 + \sin(2\theta)}\, d\theta \implies I = \frac{\pi}{4} \int_0^{\pi/2} \frac{1}{1 + \sin(2\theta)}\, d\theta$$

We evaluate the remaining elementary trigonometric integral by expressing the denominator in terms of the tangent of θ. Dividing numerator and denominator by $\cos^2\theta$:

$$\frac{1}{1 + \sin(2\theta)} = \frac{\sec^2\theta}{\sec^2\theta + 2\tan\theta} = \frac{\sec^2\theta}{1 + \tan^2\theta + 2\tan\theta} = \frac{\sec^2\theta}{(1 + \tan\theta)^2}$$

Substituting this form into the integral:

$$I = \frac{\pi}{4} \int_0^{\pi/2} \frac{\sec^2\theta}{(1 + \tan\theta)^2}\, d\theta$$

We perform the substitution $u = 1 + \tan\theta$, which gives $du = \sec^2\theta\, d\theta$. The limits of integration transform as follows:

- When $\theta = 0$: $u = 1 + \tan(0) = 1$.
- When $\theta \to \frac{\pi}{2}^-$: $u = 1 + \tan(\pi/2) \to \infty$.

The integral simplifies to a basic power-law form:

$$I = \frac{\pi}{4} \int_1^{\infty} \frac{1}{u^2}\, du = \frac{\pi}{4} \left[-\frac{1}{u}\right]_1^{\infty} = \frac{\pi}{4}(0 - (-1)) = \frac{\pi}{4}$$

The exact value of the double improper integral is:

$$\frac{\pi}{4}$$

Differentials

2.1 Medium Differential

First-Order Non-Linear Differential Equation

Let $f(x)$ satisfy the differential equation:

$$\frac{dy}{dx} + \frac{1}{x}y = xy^2$$

Given the initial condition $f(1) = \frac{1}{2}$, find the value of $f(2)$. (Assume continuity over the interval.)

Solution: We recognize the given differential equation as a Bernoulli differential equation of order $n = 2$:

$$\frac{dy}{dx} + P(x)y = Q(x)y^n$$

where $P(x) = \frac{1}{x}$, $Q(x) = x$, and $n = 2$. To linearize the equation, we divide both sides by y^2 (assuming $y \neq 0$):

$$y^{-2}\frac{dy}{dx} + \frac{1}{x}y^{-1} = x$$

We introduce the standard Bernoulli substitution $v = y^{1-n} = y^{-1} = \frac{1}{y}$. Differentiating v with respect to x using the chain rule:

$$\frac{dv}{dx} = -y^{-2}\frac{dy}{dx} \implies y^{-2}\frac{dy}{dx} = -\frac{dv}{dx}$$

Substituting v and its derivative into the transformed differential equation:

$$-\frac{dv}{dx} + \frac{1}{x}v = x \implies \frac{dv}{dx} - \frac{1}{x}v = -x$$

This is now a first-order linear ordinary differential equation for $v(x)$ in standard form, with integrating factor $\mu(x)$:

$$\mu(x) = \exp\left(\int -\frac{1}{x}\,dx\right) = \exp(-\ln|x|) = \frac{1}{x} \quad (\text{for } x > 0)$$

Multiplying the entire linear equation by the integrating factor $\mu(x) = \frac{1}{x}$:

$$\frac{1}{x}\frac{dv}{dx} - \frac{1}{x^2}v = -1$$

We recognize the left-hand side as the exact derivative of the product $(\mu(x)v)$:

$$\frac{d}{dx}\left(\frac{v}{x}\right) = -1$$

Integrating both sides with respect to x:

$$\frac{v}{x} = \int -1\,dx = -x + C$$

Multiplying through by x gives the general solution for $v(x)$:

$$v(x) = -x^2 + Cx$$

Reverting to the original dependent variable $y = \frac{1}{v}$:

$$y(x) = \frac{1}{-x^2 + Cx} = \frac{1}{Cx - x^2}$$

We now apply the given initial condition $f(1) = y(1) = \frac{1}{2}$ to determine the constant of integration C:

$$\frac{1}{2} = \frac{1}{C(1) - 1^2} = \frac{1}{C - 1}$$

Taking the reciprocal of both sides:

$$C - 1 = 2 \implies C = 3$$

Thus, the specific continuous solution to the differential equation on the domain of interest is:

$$f(x) = \frac{1}{3x - x^2}$$

To find the required value $f(2)$, we substitute $x = 2$ into our particular solution:

$$f(2) = \frac{1}{3(2) - 2^2} = \frac{1}{6 - 4} = \frac{1}{2}$$

The exact value is:

$$\frac{1}{2}$$

2.2 Hard Differential

Infinite Power Series Differential Equation

Given the differential equation:

$$(1 + \sin^2(y)) \ln(1 + x) \sum_{n=1}^{\infty} x^n \, dx = (\sin(2y) \ln(1 + \sin^2(y)) + 1) \, dy$$

If $\lim_{y \to \frac{3\pi}{2}^-} x(y) = 0$, find the constant of integration C. (Assume $|x| < 1$ for the convergence of the series.)

Solution: We begin by simplifying the infinite power series appearing on the left-hand side of the differential equation. For $|x| < 1$, the geometric series converges to a closed-form rational function:

$$\sum_{n=1}^{\infty} x^n = x + x^2 + x^3 + \cdots = x \sum_{n=0}^{\infty} x^n = \frac{x}{1 - x}$$

Substituting this closed form into the differential equation:

$$(1+\sin^2(y))\ln(1+x)\left(\frac{x}{1-x}\right)dx = (\sin(2y)\ln(1+\sin^2(y))+1)\,dy$$

We observe that the differential equation is completely separable between the variables x and y. Dividing both sides by $(1+\sin^2(y))$ separates the variables into the form:

$$\frac{x\ln(1+x)}{1-x}\,dx = \frac{\sin(2y)\ln(1+\sin^2(y))+1}{1+\sin^2(y)}\,dy$$

Let us integrate both sides independently:

$$\int \frac{x\ln(1+x)}{1-x}\,dx = \int \frac{\sin(2y)\ln(1+\sin^2(y))+1}{1+\sin^2(y)}\,dy + C$$

We first examine the right-hand side integral with respect to y, denoting it by I_y:

$$I_y = \int \frac{\sin(2y)\ln(1+\sin^2(y))}{1+\sin^2(y)}\,dy + \int \frac{1}{1+\sin^2(y)}\,dy$$

For the first term of I_y, let $u = 1+\sin^2(y)$. Differentiating gives $du = 2\sin(y)\cos(y)\,dy = \sin(2y)\,dy$. The integral becomes:

$$\int \frac{\ln(u)}{u}\,du = \frac{1}{2}(\ln(u))^2 = \frac{1}{2}\left[\ln(1+\sin^2(y))\right]^2$$

For the second term of I_y, we divide numerator and denominator by $\cos^2(y)$:

$$\int \frac{\sec^2(y)}{\sec^2(y)+\tan^2(y)}\,dy = \int \frac{\sec^2(y)}{1+2\tan^2(y)}\,dy$$

Let $v = \sqrt{2}\tan(y) \implies dv = \sqrt{2}\sec^2(y)\,dy$:

$$\frac{1}{\sqrt{2}}\int \frac{1}{1+v^2}\,dv = \frac{1}{\sqrt{2}}\tan^{-1}(v) = \frac{1}{\sqrt{2}}\tan^{-1}(\sqrt{2}\tan(y))$$

Combining both integrated terms for y:

$$I_y = \frac{1}{2}\left[\ln(1+\sin^2(y))\right]^2 + \frac{1}{\sqrt{2}}\tan^{-1}(\sqrt{2}\tan(y))$$

Now we analyze the asymptotic condition as $y \to \frac{3\pi}{2}^-$. We are given that $\lim_{y\to\frac{3\pi}{2}^-} x(y) = 0$. Let us evaluate the limit of the left-hand side integral $I_x(x) = \int_0^x \frac{t\ln(1+t)}{1-t}\,dt$ as $x \to 0$:

$$\lim_{x\to 0} I_x(x) = 0$$

Now let us evaluate the limit of the right-hand side expression $I_y(y)$ as $y \to \frac{3\pi}{2}^-$:

- As $y \to \frac{3\pi}{2}^-$, $\sin(y) \to -1 \implies \sin^2(y) \to 1$. Therefore:

$$\lim_{y\to\frac{3\pi}{2}^-} \frac{1}{2}\left[\ln(1+\sin^2(y))\right]^2 = \frac{1}{2}(\ln(2))^2$$

- As $y \to \frac{3\pi}{2}^-$ from the left (i.e., in the third quadrant where $\tan(y) > 0$), $\tan(y) \to +\infty$:
$$\lim_{y \to \frac{3\pi}{2}^-} \frac{1}{\sqrt{2}} \tan^{-1}(\sqrt{2}\tan(y)) = \frac{1}{\sqrt{2}}\left(\frac{\pi}{2}\right) = \frac{\pi}{2\sqrt{2}}$$

Equating the limits of both sides of the integrated differential equation:

$$0 = \frac{1}{2}(\ln(2))^2 + \frac{\pi}{2\sqrt{2}} + C$$

Solving explicitly for the integration constant C:

$$C = -\frac{1}{2}(\ln(2))^2 - \frac{\pi}{2\sqrt{2}}$$

The exact value of the constant of integration is:

$$-\frac{1}{2}(\ln(2))^2 - \frac{\pi}{2\sqrt{2}}$$

Derivatives

3.1 Medium Derivative

Lambert W Function Composite Derivative

Let $W(x)$ denote the Lambert W function, defined implicitly by $W(x)e^{W(x)} = x$. Find the derivative of

$$f(x) = \sin(W(x)) \quad \text{at } x = e^{\frac{\pi}{6}+\ln\left(\frac{\pi}{6}\right)}.$$

Solution: We begin by simplifying the evaluation point $x_0 = e^{\frac{\pi}{6}+\ln\left(\frac{\pi}{6}\right)}$ using exponential and logarithmic identities:

$$x_0 = e^{\pi/6} \cdot e^{\ln(\pi/6)} = \frac{\pi}{6}e^{\pi/6}$$

By definition of the Lambert W function, $W(z)$ is the inverse relation of $g(w) = we^{w}$. We observe that our evaluation coordinate x_0 is already expressed explicitly in the form $w_0e^{w_0}$ with $w_0 = \frac{\pi}{6}$. Therefore, applying the Lambert W function to x_0 yields:

$$W(x_0) = W\left(\frac{\pi}{6}e^{\pi/6}\right) = \frac{\pi}{6}$$

We now differentiate the composite function $f(x) = \sin(W(x))$ with respect to x using the chain rule:

$$f'(x) = \cos(W(x)) \cdot W'(x)$$

To find the general derivative $W'(x)$ of the Lambert W function, we differentiate the defining implicit relation $W(x)e^{W(x)} = x$ with respect to x using the product and chain rules:

$$W'(x)e^{W(x)} + W(x)e^{W(x)}W'(x) = 1$$

Factoring out $W'(x)e^{W(x)}$ on the left-hand side:

$$W'(x)e^{W(x)}(1 + W(x)) = 1 \implies W'(x) = \frac{1}{e^{W(x)}(1 + W(x))}$$

Since the defining relation gives $e^{W(x)} = \frac{x}{W(x)}$ for $x \neq 0$, we can also write this derivative equivalently as:

$$W'(x) = \frac{W(x)}{x(1 + W(x))}$$

Using the exponential form of $W'(x)$, we substitute into our expression for $f'(x)$:

$$f'(x) = \frac{\cos(W(x))}{e^{W(x)}(1 + W(x))}$$

We evaluate this derivative at the target point x_0, where $W(x_0) = \frac{\pi}{6}$:

$$f'(x_0) = \frac{\cos\left(\frac{\pi}{6}\right)}{e^{\pi/6}\left(1 + \frac{\pi}{6}\right)}$$

We substitute the exact trigonometric value $\cos\left(\frac{\pi}{6}\right) = \frac{\sqrt{3}}{2}$ and simplify the algebraic fraction in the denominator:

$$1 + \frac{\pi}{6} = \frac{6+\pi}{6}$$

$$f'(x_0) = \frac{\frac{\sqrt{3}}{2}}{e^{\pi/6}\left(\frac{6+\pi}{6}\right)} = \frac{\sqrt{3}}{2} \cdot \frac{6}{(6+\pi)e^{\pi/6}} = \frac{3\sqrt{3}}{(6+\pi)e^{\pi/6}}$$

The exact value of the derivative is:

$$\frac{3\sqrt{3}}{(6+\pi)e^{\pi/6}}$$

3.2 Hard Derivative

Nested Radical Inverse Function Derivative

Let $(f^{-1}(x))'$ denote the derivative of the inverse function. If $f(x) = \sqrt{x} + \sqrt{x+1}$, find the value of

$$\left(f^{-1}\left((1+\sqrt{2})^2\right)\right)'.$$

Solution: Let $y = f(x) = \sqrt{x} + \sqrt{x+1}$ for $x \geq 0$. We first derive an explicit analytical formula for the inverse function $x = f^{-1}(y)$. Notice that the conjugate radical expression satisfies:

$$(\sqrt{x+1} + \sqrt{x})(\sqrt{x+1} - \sqrt{x}) = (x+1) - x = 1$$

Since $y = \sqrt{x+1} + \sqrt{x}$, taking the reciprocal yields:

$$\frac{1}{y} = \sqrt{x+1} - \sqrt{x}$$

We now have a system of two linear equations in terms of the radicals $\sqrt{x+1}$ and $\sqrt{x}$:

$$\begin{aligned} \sqrt{x+1} + \sqrt{x} &= y \\ \sqrt{x+1} - \sqrt{x} &= \frac{1}{y} \end{aligned}$$

Subtracting the second equation from the first eliminates $\sqrt{x+1}$:

$$2\sqrt{x} = y - \frac{1}{y} \implies \sqrt{x} = \frac{1}{2}\left(y - \frac{1}{y}\right)$$

Squaring both sides expresses x explicitly as a function of y:

$$x = f^{-1}(y) = \frac{1}{4}\left(y - \frac{1}{y}\right)^2 = \frac{1}{4}\left(y^2 - 2 + \frac{1}{y^2}\right)$$

To find the derivative of the inverse function $(f^{-1}(y))'$, we differentiate $f^{-1}(y)$ with respect to y:

$$\left(f^{-1}(y)\right)' = \frac{d}{dy}\left[\frac{1}{4}\left(y^2 - 2 + y^{-2}\right)\right] = \frac{1}{4}\left(2y - 2y^{-3}\right) = \frac{1}{2}\left(y - \frac{1}{y^3}\right)$$

We are required to evaluate this derivative at the specific point $y_0 = (1+\sqrt{2})^2$. Let us expand y_0:

$$y_0 = (1+\sqrt{2})^2 = 1 + 2\sqrt{2} + 2 = 3 + 2\sqrt{2}$$

We also compute the reciprocal $\frac{1}{y_0}$ by rationalizing the denominator:

$$\frac{1}{y_0} = \frac{1}{3+2\sqrt{2}} \cdot \frac{3-2\sqrt{2}}{3-2\sqrt{2}} = \frac{3-2\sqrt{2}}{9-8} = 3 - 2\sqrt{2}$$

We need the cubed reciprocal $\frac{1}{y_0^3} = \left(\frac{1}{y_0}\right)^3$:

$$\frac{1}{y_0^3} = (3-2\sqrt{2})^3$$

Using the binomial expansion $(a-b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$ with $a = 3$ and $b = 2\sqrt{2}$:

$$(3-2\sqrt{2})^3 = 3^3 - 3(3^2)(2\sqrt{2}) + 3(3)(2\sqrt{2})^2 - (2\sqrt{2})^3$$
$$= 27 - 54\sqrt{2} + 9(8) - 16\sqrt{2} = 27 - 54\sqrt{2} + 72 - 16\sqrt{2} = 99 - 70\sqrt{2}$$

Now we substitute $y_0 = 3 + 2\sqrt{2}$ and $y_0^{-3} = 99 - 70\sqrt{2}$ into our derivative formula:

$$\left(f^{-1}(y_0)\right)' = \frac{1}{2}\left(y_0 - \frac{1}{y_0^3}\right) = \frac{1}{2}\left[(3+2\sqrt{2}) - (99-70\sqrt{2})\right]$$

$$= \frac{1}{2}\left(3 + 2\sqrt{2} - 99 + 70\sqrt{2}\right) = \frac{1}{2}\left(-96 + 72\sqrt{2}\right) = -48 + 36\sqrt{2}$$

The exact value of the derivative is:

$$36\sqrt{2} - 48$$

Physics & Engineering

4.1 Mechanics

Elliptical Rail Work-Energy Calculation

An elliptically shaped rail PQ lies in the vertical plane with $OP = 3$ m and $OQ = 4$ m. A block of mass 1 kg is pulled from P to Q by a constant force of 18 N which is always parallel to the line PQ. Assuming no friction, find the kinetic energy (in J) at Q. Take $g = 10$ m s^{-2}.

Solution: Let us establish a Cartesian coordinate system with origin $O(0, 0)$ at the center of the ellipse. The rail PQ lies in the vertical plane representing a quadrant of an ellipse. Without loss of generality, let the point P be situated on the positive horizontal axis at coordinates $P(3, 0)$ and the point Q be situated on the positive vertical axis at coordinates $Q(0, 4)$, so that the semi-axes are $a = OP = 3$ m horizontally and $b = OQ = 4$ m vertically.

A block of mass $m = 1$ kg is initially at rest at point P and slides along the frictionless elliptical rail to point Q. By the Work-Energy Theorem, the change in kinetic energy $\Delta K = K_Q - K_P$ of the block is equal to the total net work W_{net} done by all forces acting on the block along the trajectory from P to Q:

$$K_Q - K_P = W_{\text{net}} = W_F + W_g + W_N$$

Since the block starts from rest at P, the initial kinetic energy is $K_P = 0$, so $K_Q = W_{\text{net}}$. We analyze the work done by each individual force:

1. **Normal Force (W_N):** Because the rail is completely frictionless, the normal contact force $\mathbf{N}$ exerted by the rail is always perpendicular to the instantaneous velocity vector $d\mathbf{r}$ of the block. Therefore, the normal force does zero work:

$$W_N = \int_P^Q \mathbf{N} \cdot d\mathbf{r} = 0$$

2. **Gravitational Force (W_g):** Gravity acts vertically downward with constant magnitude mg. As the block moves from $P(3, 0)$ to $Q(0, 4)$, its vertical elevation increases by $\Delta y = y_Q - y_P = 4 - 0 = 4$ m. The work done by gravity is equal to the negative of the change in gravitational potential energy:

$$W_g = -mg\Delta y = -(1\text{ kg})(10\text{ m s}^{-2})(4\text{ m}) = -40\text{ J}$$

3. **Applied Constant Force (W_F):** The pulling force $\mathbf{F}$ has a constant magnitude of $F = 18$ N and maintains a fixed direction parallel to the straight secant line connecting $P(3, 0)$ to $Q(0, 4)$. The displacement vector between the endpoints of the motion is:

$$\Delta\mathbf{r} = \mathbf{r}_Q - \mathbf{r}_P = (0 - 3)\hat{\mathbf{i}} + (4 - 0)\hat{\mathbf{j}} = -3\hat{\mathbf{i}} + 4\hat{\mathbf{j}}$$

The straight-line distance between P and Q is:

$$d = \|\Delta\mathbf{r}\| = \sqrt{(-3)^2 + 4^2} = \sqrt{9 + 16} = 5\text{ m}$$

Since the force vector **F** is constant in both magnitude and direction along the entire curvilinear path, the work done by **F** depends only on the total vector displacement $\Delta\mathbf{r}$:

$$W_F = \int_P^Q \mathbf{F} \cdot d\mathbf{r} = \mathbf{F} \cdot \int_P^Q d\mathbf{r} = \mathbf{F} \cdot \Delta\mathbf{r}$$

Because **F** is directed parallel to the line PQ (in the direction from P to Q), the angle between **F** and $\Delta\mathbf{r}$ is zero ($\cos 0^\circ = 1$). Thus:

$$W_F = F\|\Delta\mathbf{r}\| = (18\text{ N})(5\text{ m}) = 90\text{ J}$$

Summing the work contributions to find the final kinetic energy at point Q:

$$K_Q = W_F + W_g + W_N = 90\text{ J} - 40\text{ J} + 0\text{ J} = 50\text{ J}$$

The exact kinetic energy of the block at Q is:

$$50\text{ J}$$

4.2 Quantum Mechanics

Relativistic Electron de Broglie Wavelength

An electron is accelerated through a potential difference of $V = 2.50$ MV. Assuming relativistic effects are significant, calculate its de Broglie wavelength. Express the answer in units of 10^{-13} m. Use $h = 6.626 \times 10^{-34}$ J s, $m_e c^2 = 0.511$ MeV and $c = 3.00 \times 10^8$ m s^{-1}.

Solution: When an electron of elementary charge e is accelerated from rest through an electrostatic potential difference $V = 2.50$ MV, it acquires a relativistic kinetic energy K given by:

$$K = eV = 2.50\text{ MeV}$$

In relativistic quantum mechanics, the total energy E of the particle is the sum of its rest mass energy $E_0 = m_e c^2$ and its kinetic energy K:

$$E = K + E_0 = 2.50\text{ MeV} + 0.511\text{ MeV} = 3.011\text{ MeV}$$

The relativistic energy-momentum dispersion relation is:

$$E^2 = (pc)^2 + E_0^2 \implies pc = \sqrt{E^2 - E_0^2}$$

Substituting our values for E and E_0:

$$pc = \sqrt{(3.011)^2 - (0.511)^2}\text{ MeV}$$

We compute the squares:

$$(3.011)^2 = 9.066121, \qquad (0.511)^2 = 0.261121$$

Subtracting the two quantities:

$$E^2 - E_0^2 = 9.066121 - 0.261121 = 8.805000\ \text{MeV}^2$$

Taking the square root gives the momentum-light speed product:

$$pc = \sqrt{8.805} \approx 2.967322\ \text{MeV}$$

According to the de Broglie hypothesis, the wavelength λ of a matter wave is related to the particle momentum p by $\lambda = \frac{h}{p}$. Multiplying numerator and denominator by the speed of light c:

$$\lambda = \frac{hc}{pc}$$

We convert the product hc from SI units (J s $\cdot$ m s^{-1}) into MeV m:

$$hc = (6.626 \times 10^{-34}\ \text{J s})(3.00 \times 10^8\ \text{m s}^{-1}) = 1.9878 \times 10^{-25}\ \text{J m}$$

Since $1\ \text{eV} = 1.60217663 \times 10^{-19}\ \text{J} \implies 1\ \text{MeV} = 1.60217663 \times 10^{-13}\ \text{J}$:

$$hc = \frac{1.9878 \times 10^{-25}\ \text{J m}}{1.60217663 \times 10^{-13}\ \text{J/MeV}} \approx 1.240687 \times 10^{-12}\ \text{MeV m} = 1.240687 \times 10^{-6}\ \text{MeV m}$$

$$\text{(or } 1240.687\ \text{MeV fm)}$$

Using SI conversions directly for precision, let us convert $pc = 2.967322$ MeV into Joules:

$$pc = (2.967322 \times 10^6\ \text{eV})(1.60217663 \times 10^{-19}\ \text{J/eV}) \approx 4.754173 \times 10^{-13}\ \text{J}$$

Thus, the de Broglie wavelength is:

$$\lambda = \frac{hc}{pc} = \frac{1.9878 \times 10^{-25}\ \text{J m}}{4.754173 \times 10^{-13}\ \text{J}} \approx 4.181168 \times 10^{-13}\ \text{m}$$

Rounding to three significant figures to match the input data precision ($V = 2.50$ MV, $c = 3.00 \times 10^8$ m s^{-1}):

$$\lambda \approx 4.18 \times 10^{-13}\ \text{m}$$

Expressed in units of 10^{-13} m, the numerical value is:

$$4.18$$

4.3 Electromagnetism

Cube Resistor Network Body Diagonal Current

A cube-shaped resistor network has eight vertices and twelve edges. Each edge contains an identical ideal resistor of resistance $R = 12\,\Omega$. An ideal $V = 10$ V battery is connected between two opposite vertices of the cube (a body diagonal). Determine the total current supplied by the battery in amperes.

Solution: Let us label the two opposite body-diagonal vertices connected to the external battery as node A (current entry point at potential $V_A = V = 10$ V) and node B (current exit point at reference potential $V_B = 0$ V). To determine the total current I_{total} supplied by the battery, we first calculate the equivalent resistance R_{eq} of the twelve-edge cubic skeleton between body-diagonal vertices A and B.

We exploit the symmetry of the cubic lattice by partitioning the eight vertices into four distinct equipotential sets based on their shortest path distance (number of edges) from the entry node A:

1. **Set 0 (Distance 0):** The entry node A itself (1 vertex).
2. **Set 1 (Distance 1):** The three vertices directly adjacent to A. By threefold rotational symmetry around the body diagonal AB, these three vertices are at the exact same electric potential V_1.
3. **Set 2 (Distance 2):** The three vertices directly adjacent to the exit node B (distance 2 from A). By symmetry, these three vertices are also at an identical electric potential V_2.
4. **Set 3 (Distance 3):** The exit node B itself (1 vertex).

Because vertices within Set 1 are at the same potential V_1, no current would flow through any imaginary short-circuit wire connecting them; therefore, we can connect (node-condense) the three vertices of Set 1 into a single equipotential node. Similarly, we condense the three vertices of Set 2 into a single equipotential node. This condensation groups the twelve edges of the cube into three distinct series-connected stages between node A and node B:

- **Stage 1 (From Set 0 to Set 1):** There are 3 edges connecting node A to the three adjacent vertices in Set 1. Since these 3 identical resistors of resistance R are in parallel, their equivalent resistance is:

$$R_{\text{stage 1}} = \frac{R}{3}$$

- **Stage 2 (From Set 1 to Set 2):** Each of the 3 vertices in Set 1 connects to 2 distinct vertices in Set 2, accounting for $3 \times 2 = 6$ edges in total. These 6 identical resistors are connected in parallel between the condensed Set 1 node and the condensed Set 2 node:

$$R_{\text{stage 2}} = \frac{R}{6}$$

- **Stage 3 (From Set 2 to Set 3):** There are 3 edges connecting the three vertices in Set 2 to the exit node B. These 3 identical resistors are in parallel:

$$R_{\text{stage 3}} = \frac{R}{3}$$

Summing the equivalent resistances of the three series stages yields the classic equivalent resistance R_{eq} across the body diagonal of a cube:

$$R_{\text{eq}} = R_{\text{stage 1}} + R_{\text{stage 2}} + R_{\text{stage 3}} = \frac{R}{3} + \frac{R}{6} + \frac{R}{3} = \frac{2R + R + 2R}{6} = \frac{5}{6}R$$

Substituting the given edge resistance $R = 12\,\Omega$:

$$R_{\text{eq}} = \frac{5}{6}(12\,\Omega) = 10\,\Omega$$

By Ohm's Law, the total current I_{total} supplied by the $V = 10\,\text{V}$ battery to the resistor network is:

$$I_{\text{total}} = \frac{V}{R_{\text{eq}}} = \frac{10\text{ V}}{10\,\Omega} = 1\text{ A}$$

The total current supplied by the battery is:

$$1\text{ A}$$